

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 4: Quadratic Equations

Previous year board questions, topic-wise: 8 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Word problems

1. Over a 200 km journey, a fast train takes one hour less than a slow train. The slow train runs 10 km/h slower than the fast one. Find the speed of each train. **CBSE 2026, 5 marks**
2. Two squares have a total area of 640 m², and their perimeters differ by 64 m. Find the side of each square. **CBSE 2026, 5 marks**
3. A right-angled triangle has a perimeter of 60 cm and a hypotenuse of 25 cm. Find the lengths of the other two sides. **CBSE 2025, 5 marks**
4. A train covers 480 km at a uniform speed. Had it travelled 8 km/h slower, the journey would have taken 3 hours longer. Find the speed of the train. **CBSE 2025, 5 marks**
5. A rectangular floor is covered exactly by 200 square tiles. With tiles whose side is 1 unit longer, only 128 tiles are needed. Let the side of an original tile be x units. (i) Form a quadratic equation in x . (ii) Write it in standard form. (iii) Find x by factorisation, OR solve the equation by the quadratic formula. **CBSE 2024, 4 marks, case study**
6. A school photo is 18 cm long and 12 cm wide. Its length and width are each to be increased by x cm so that the area doubles. (i) Write an equation for this. (ii) Write it as a quadratic equation in standard form. (iii) Find the new length and width, OR decide whether any rational value of x gives a new area of 220 cm². **CBSE 2023, 4 marks**

2 Nature of roots

7. The equation $ax^2 + bx + c = 0$, $a \neq 0$, has real and equal roots. Which relation holds? (a) $a = \frac{b^2}{c}$ (b) $b^2 = ac$ (c) $ac = \frac{b^2}{4}$ (d) $c = \frac{b^2}{a}$ **CBSE 2024, 1 mark**
8. What is the least positive value of k for which $2x^2 + kx - 4 = 0$ has rational roots? (a) $\pm 2\sqrt{2}$ (b) 2 (c) ± 2 (d) $\sqrt{2}$ **CBSE 2023, 1 mark**

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